Teamwork Assignment 2 (maximum 4 people)

Deadline: 20 June 2022

This project aims at modelling the first stage of an actual launch vehicle. To this end, the following is requested:

- Choose a particular launch vehicle system (not necessarily a recent design), and find as much information as possible of it, specifically:
  - General description.
  - Distribution of masses: structure, propellant (of each stage), and payload.
  - Fuel and oxidizer, including their proportions (only for the first stage).
  - Combustor pressure p_c and temperature T_c (only for the first stage).

- Find a mathematical model of the atmosphere according to the particular range of altitudes of the first stage.

- Use a thermochemistry software (e.g., CEA) and find the combustor temperature T_c, as well as the molecular mass M of the gas (assume frozen composition) and a suitable mean value of the adiabatic coefficient γ.

- Taking the nominal area ratio A_e/A_t, determine the throat area $A_t$necessary to produce the initial thrust at sea level, and find the mass flow $\dot{m}$ and the burning time t_b, and determine the altitude and velocity vs. time, assuming either a vertical or oblique ascent.

- Resize the nozzle and find a new area ratio A_e/A_tthat improves the performance in terms of impulse $\Delta V = V_f - V_0$ and gain of altitude Δh, assuming a weight penalty of structural mass proportional to the exit area A_e, according to

$$\Delta m_{st} = m_0 \frac{\Delta A_e}{A_t} \varepsilon \quad \text{with}$$

  ε being a parameter, left free for each team, in the range $0.00005 < \varepsilon < 0.0001$.

Remarks:

- Variation of gravity with altitude

$$g = \frac{g_0}{\left(1 + \frac{h}{R_T}\right)^2}$$

- If an oblique launch is considered, the student team may (optionally) consider thrust vectoring, as well as investigate (in the literature) possible models for both lift and drag coefficients. Alternatively, for either vertical or oblique launch, in absence of other sources, the students may use the drag (and lift) coefficient(s) given in the annex (referred to the reference cross sectional area S_{ref}).

ANNEXES

Annex 1. Aerodynamic drag coefficient for zero lift (vertical ascent)

Drag from dynamic pressure, reference cross sectional area and drag coefficient

$$D = \frac{1}{2}\rho V^2 S_{ref} C_D$$

$M \leq 0.6 \rightarrow \ C_D = 0.15$

$0.6 < M \leq 1.00 \ \rightarrow \ C_D = -1.17188M^3 \ + \ 3.51563\,M^2 - 2.95313\,M \ + \ 0.90938$

$1.00 < M \leq 1.18 \rightarrow C_D = -23.79115M^3 + 76.23457M^2 - 80.53318M + 28.38976$

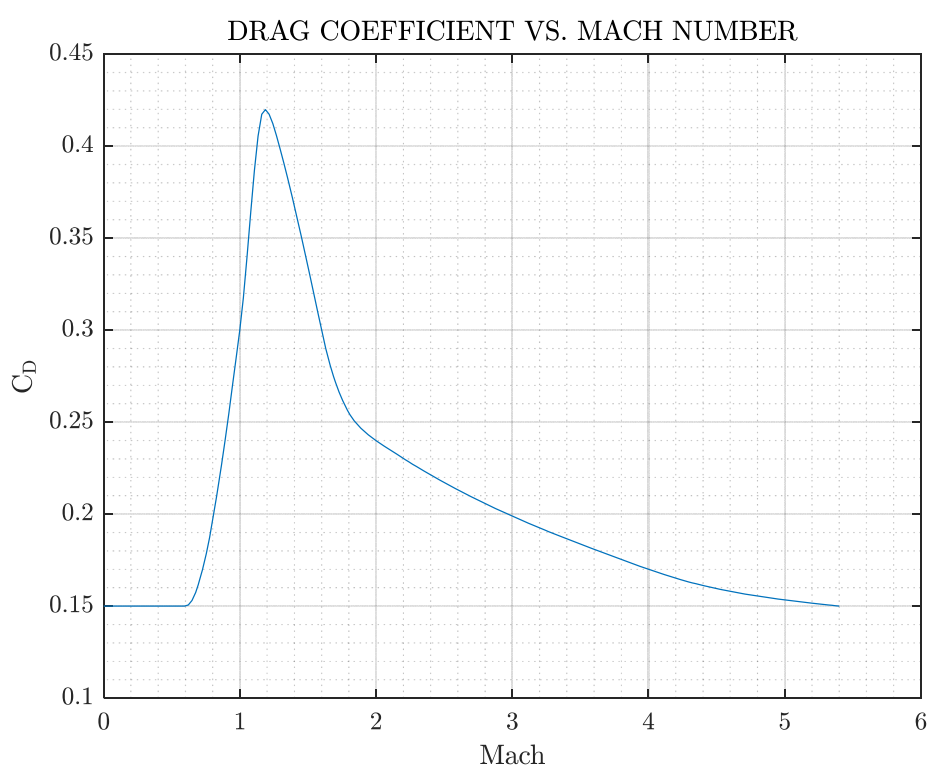
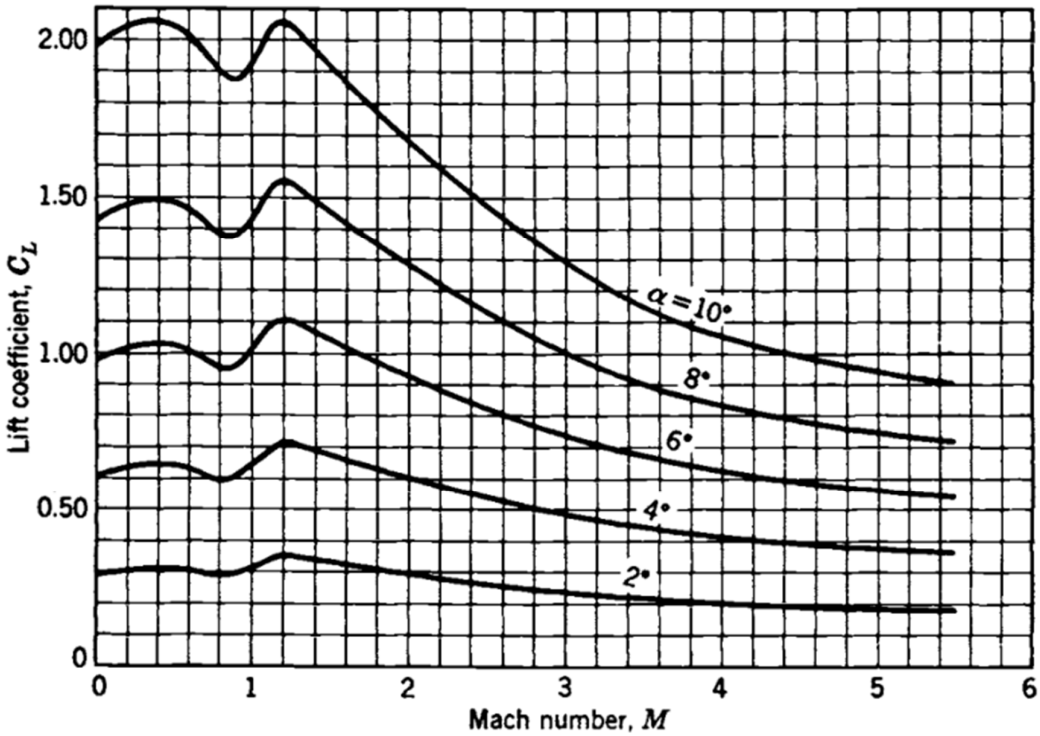
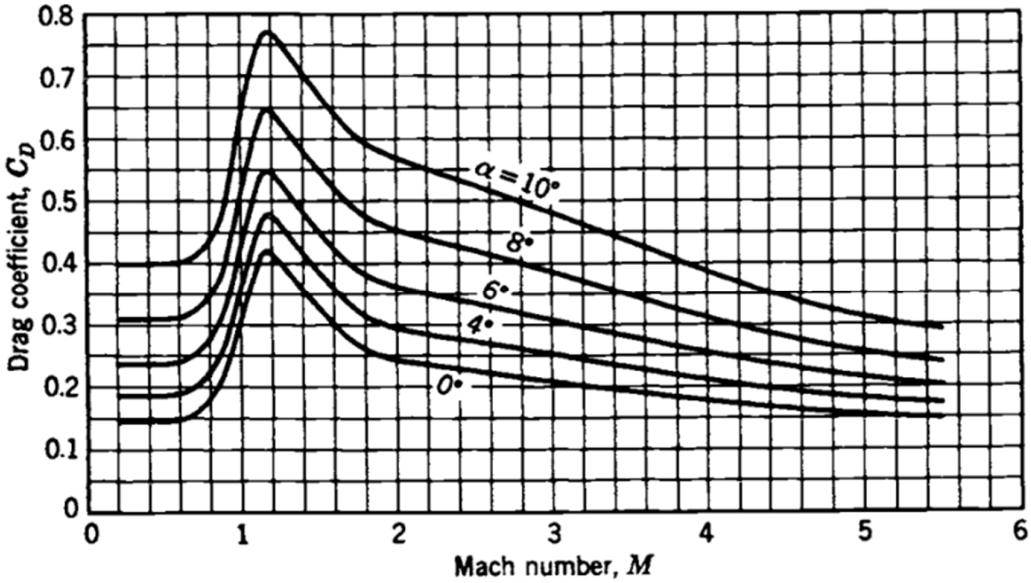
$1.18 < M \leq 1.29 \rightarrow \ C_D = 7.51315M^3 - 29.07588\,M^2 + 37.23516M - 15.37657$

$1.29 < M \leq 1.60 \rightarrow \ C_D = 0.25938M^3 - 1.24504M^2 + 1.64455M - 0.20642$

$1.60 < M \leq 2.00 \rightarrow \ C_D = -0.61721M^3 + \ 3.70326M^2 - 7.45777M + 5.28018$

$2.00 < M \leq 4.00 \rightarrow \ C_D = -0.00203M^3 + \ 0.02437M^2 - 0.12435M + \ 0.40748$

$4.00 < M \leq 5.40 \rightarrow \ C_D = -0.00321M^3 + \ 0.05203M^2 - 0.28898M + \ 0.69892$



Annex 2. Aerodynamic lift and drag coefficients